

Vulnerability assessment

OpenVAS/Greenbone

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Chapter 1

OpenVAS Scan

In this chapter the most important aspects of OpenVAS are explained, with practical examples to give the reader an understanding of the tools usage.

1.1 OpenVAS Scan

In this section it will be explained how to start a scan (that could be just for an host, or a group of hosts). We will create a target, which means declaring which host (or group of hosts) to be scanned, which port range and protocol.

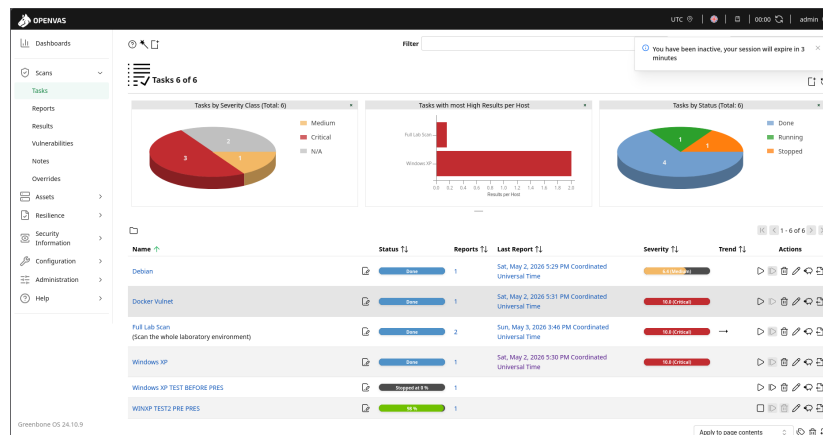
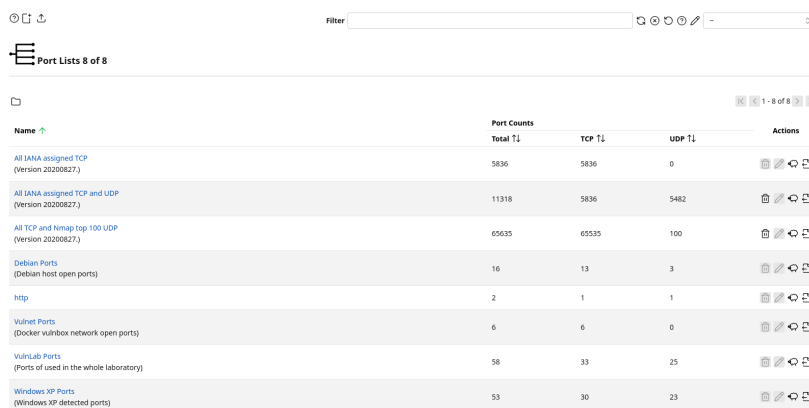


Figure 1.1: Scan windows

This is the scan windows, where you will create the tasks which are configurations for the targets and ports that are going to be scanned, how many NVTs (Network Vulnerability Test) are going to be used for the scan, how many concurrent hosts are being scanned at a given moment etc. We will get to all of the configuration in time.

1.1.1 Configuration → Ports



Port Lists 8 of 8

Name	Port Counts			Actions
	Total T1	TCP T1	UDP T1	
All IANA assigned TCP (Version 20200827)	5836	5836	0	[Add] [Edit] [Delete]
All IANA assigned TCP and UDP (Version 20200827)	11318	5836	5482	[Add] [Edit] [Delete]
All TCP and Nmap top 100 UDP (Version 20200827)	65635	65635	100	[Add] [Edit] [Delete]
Debian Ports (Debian host open ports)	16	13	3	[Add] [Edit] [Delete]
http	2	1	1	[Add] [Edit] [Delete]
Vulnet Ports (Docker vulnbox network open ports)	6	6	0	[Add] [Edit] [Delete]
VulnLab Ports (Ports of used in the whole laboratory)	58	33	25	[Add] [Edit] [Delete]
Windows XP Ports (Windows XP detected ports)	53	30	23	[Add] [Edit] [Delete]

Figure 1.2: Port creation interface

This is the ports creation interface, you can create custom ports ranges looking for services that use TCP/UDP. For the most part OpenVAS works by giving us templates, in this case are for a complete scans of all IANA (Internet Assigned Numbers Authority) ports, which means it will scan for all 65536 ports only TCP, UDP, both etc. It really just depends on which preset is chosen.

OpenVAS encodes ports in a format like T:1 – 1023, 4500,U:2 – 100, 3246 which means that it will scan for TCP ports between 1 and 1023 and 4500 alone, and for UDP ports from 2 to 100 and 3246 alone. In this way you can define ports range and choose to do only TCP or UDP scans.

Under the hood OpenVAS uses Nmap to scan the ports, and just like it to scan UDP ports it will take considerably longer than TCP. This is because when a TCP request is made, the server either accept the connection or it rejects it, while for UDP being a connection-less protocol will wait until timeout. So beware about this aspect if you don't want the scan to take hours.

For our experiments we will create two presets: one for ftp service (port 21) and one for an http server (port 80). Go on the add port icon on the top of the interface (the little square with a plus), and you should get this popup.

New Port List

Name

WIN XP ONLY FTP

Comment

Port Ranges

☒ Manual

T:21

☐ From file

Cancel

Save

New Port List

Name

WIN XP FTP&HTTP

Comment

Port Ranges

☒ Manual

T:21,80

☐ From file

Cancel

Save

(a) Port configuration limited to TCP 21 (FTP). (b) Port configuration limited to TCP 21 and 80 (FTP).

Figure 1.3: Ports configuration

Those configurations are going to be used later to scan a Windows XP target with FTP and HTTP servers.

Credentials for target access

The screenshot shows the OpenVAS web interface with the 'Credentials' window open. The window displays a table of 10 credentials. The table has columns for Name, Type, Allow insecure use, Login, and Actions. The credentials listed are: Admin, Admin/vulnbox, Debian, Guest, Manager, MySQL/vulnbox, PostgreSQL, Root, Tomcat, and User. Each credential is of type 'Username + Password' and has 'Yes' for 'Allow insecure use'. The 'Login' column shows the username for each credential. The 'Actions' column contains icons for editing, deleting, and other operations.

Name	Type	Allow insecure use	Login	Actions
Admin	Username + Password	Yes	admin	[Edit] [Delete] [Add]
Admin/vulnbox	Username + Password	Yes	admin	[Edit] [Delete] [Add]
Debian	Username + Password	Yes	debian	[Edit] [Delete] [Add]
Guest	Username + Password	Yes	guest	[Edit] [Delete] [Add]
Manager	Username + Password	Yes	manager	[Edit] [Delete] [Add]
MySQL/vulnbox	Username + Password	Yes	vulnbox	[Edit] [Delete] [Add]
Postgres	Username + Password	Yes	postgres	[Edit] [Delete] [Add]
Root	Username + Password	Yes	root	[Edit] [Delete] [Add]
Tomcat	Username + Password	Yes	tomcat	[Edit] [Delete] [Add]
User	Username + Password	Yes	user	[Edit] [Delete] [Add]

Figure 1.4: Credentials window

Credentials are used for deeper scans, you give the credentials to OpenVAS and it will try to login to the machine, scanning the host from within, looking at configuration files, installed packages etc. This is done to improve scan performance since it flags problems that are not visible from the outside, but may lead to exploitable vulnerabilities. This function can also be used to check for default credentials if they are in use or no.

New Credential

×

Name

Test

Comment

Type

Username + Password

Allow insecure use

☐ Yes
☒ No

Auto-generate

☐ Yes
☒ No

Username

Password

Cancel

Save

New Credential

×

Name

Test

Comment

Type

Username + Password

Username + SSH Key

SNMP

S/MIME Certificate

PGP Encryption Key

Password only

Client Certificate

Cancel

Save

(a) FTP Target Step 1

(b) FTP Target Step 2

Figure 1.5: Configuration for the FTP only target

As you can see many possibilities are available, setting both username and password, ssh access, certificates etc. But for our example this feature will not be used.

1.1.2 Target configuration

Dashboards

Scans

Assets

Resilience

Security Information

Configuration

Targets

Port Lists

Credentials

Scan Configs

Report Configs

Report Formats

Scanners

Filters

Tags

Administration

Help

Filter

Targets 13 of 13

Name	Hosts T1	IPs T1	Port List T1	Credentials	Actions
Debian Target (Debian VM vulnbox)	192.168.56.103	1	Debian Ports	SSH:Debian SSH:Elevate-Root SMB:NTLMAuth/Debian	
VM Targets (VMs in the laboratory environment)	192.168.56.101-192.168.56.104	4	VulnLab Ports	SSH:Debian SSH:Elevate-Root SMB:NTLMAuth/Debian	
vulnbox1 (Log4Shell)	172.30.0.2	1	VulnLab Ports	SSH:Admin SSH:Elevate-Root	
vulnbox2 (Apache Struts)	172.30.1.2	1	VulnLab Ports	SSH:Tomcat SSH:Elevate-Root	
vulnbox3 (CVE-2019-10243)	172.30.2.2	1	VulnLab Ports	SSH:MySQL/vulnbox3 SSH:Elevate-Root	
vulnbox4 (SSRF)	172.30.2.3	1	VulnLab Ports	SSH:User SSH:Elevate-Admin	
vulnbox5 (Samba)	172.30.2.4	1	VulnLab Ports	SSH:Guest SSH:Elevate-Root SMB:NTLMAuth/Guest	

Figure 1.6: Target configuration interface

This is the target creation interface, which is used to create all targets that are going to be scanned expressed in single IP or a range of IPs (e.g. 192.168.56.101-192.168.56.105).

Before scanning the Windows XP server you would want to go to the terminal of the machine and type **ipconfig /all**, copy the IP (which should be along the lines of 192.168.56.10X).

```

C:\Documents and Settings\windows>ipconfig /all

Windows IP Configuration

Host Name . . . . . : windows-xp
Primary Dns Suffix . . . . . :
Node Type . . . . . : Unknown
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) PRO/1000 T Server Adapter
Physical Address. . . . . : 08-00-27-BB-85-A1
Dhcp Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IP Address. . . . . : 192.168.56.104
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCP Server . . . . . : 192.168.56.100
Lease Obtained. . . . . : Tuesday, May 12, 2026 2:35:15 PM
Lease Expires . . . . . : Tuesday, May 12, 2026 2:45:15 PM

C:\Documents and Settings\windows>

```

Figure 1.7: Terminal output of ipconfig command

After the IP was discovered, its time to configure the target like so:

(a) FTP Target Step 1

(b) FTP Target Step 2

Figure 1.8: Configuration for the FTP only target

In this popup you will set target name, the IP (or range of IPs), the excluded hosts in case you have a range of IPs and some host are not to be scanned, the port list preset made before and the credentials presets that are set in the credentials window (not used in this example).

Now configure the targets also for the FTP and HTTP target like so:

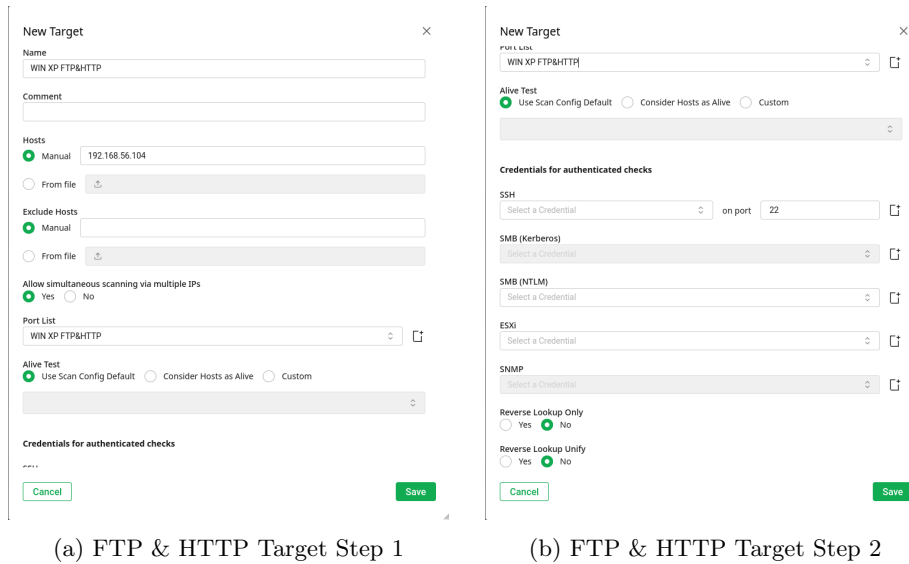


Figure 1.9: Configuration for the FTP & HTTP target

Scan configs

Name	Family	Total T1	Trend T1	NVTs	Trend T1	Actions
Base (Basic configuration template with a minimum set of NVTs required for a scan. Version 20200827)	2	→	3	→		
Discovery (Network Discovery scan configuration. Version 20201215)	11	→	320	→		
Empty (Empty and static configuration template. Version 20201215)	0	→	0	→		
Full and Fast (Most NVTs, optimized by using previously collected information. Version 20201215)	60	→	176180	→		
Full and Fast Clone 1 (Most NVTs, optimized by using previously collected information. Version 20201215)	42	→	15650	→		
Full and Fast Clone 1 Clone 1 (Most NVTs, optimized by using previously collected information. Version 20201215)	42	→	15650	→		
Host Discovery (Network Host Discovery scan configuration. Version 20201215)	2	→	2	→		
Host Discovery Clone 1 (Network Host Discovery scan configuration. Version 20201215)	2	→	2	→		
Log2kill (Configuration with checks for Log2kill and CVE-2021-46228. Version 20211227)	10	→	29	→		

Figure 1.10: Scan configs window

The scan configuration is a configuration used during the scan, essentially it tells OpenVAS how to scan and what to scan. Essentially the Full and Fast preset looks at everything possible from Cisco appliances, Fedora security, Windows services etc. This is the reason why the database is so big, occupying tens of gigabytes. Since there are thousands of settings if we look at every possible configuration, what we will do is to configure a pre-set at a high level tailoring it for Windows XP.

New Scan Config

×

Name

Windows XP Taylored Scan

Comment

Base

☐ Base with a minimum set of NVTs
 ☒ Empty, static and fast
 ☐ Full and fast

CVE

Cancel

Save

Figure 1.11: Taylored scan creation

To create it select the Empty config, save it and then edit it by clicking to the pencil icon, select the following:

Edit Scan Config Windows XP Taylored Scan

×

Name

Windows XP Taylored Scan

Comment

Empty and static configuration template. Version 20201215.

Search

Search for families, preferences, or NVTs

Edit Network Vulnerability Test Families (60)

⊞

Family	NVTs selected	Trend	Select all NVTs	Actions
ADK Local Security Checks	0 of 1	⊞ ⊞ ⊞ ⊞	☒	
Amazon Linux Local Security Checks	0 of 748	⊞ ⊞ ⊞ ⊞	☐	
Brute force attacks	0 of 9	⊞ ⊞ ⊞ ⊞	☐	
Buffer overflow	0 of 682	⊞ ⊞ ⊞ ⊞	☐	
CISCO	0 of 651	⊞ ⊞ ⊞ ⊞	☐	
CentOS Local Security Checks	0 of 3255	⊞ ⊞ ⊞ ⊞	☐	
Citrix XenServer Local Security Checks	0 of 30	⊞ ⊞ ⊞ ⊞	☐	
Compliance	0 of 16	⊞ ⊞ ⊞ ⊞	☐	

Cancel

Save

Edit Scan Config Windows XP Taylored Scan

×

CISCO

0 of 651

⊞ ⊞ ⊞ ⊞

☒

CentOS Local Security Checks

0 of 3255

⊞ ⊞ ⊞ ⊞

☐

Citrix XenServer Local Security Checks

0 of 30

⊞ ⊞ ⊞ ⊞

☐

Compliance

0 of 16

⊞ ⊞ ⊞ ⊞

☐

Databases

0 of 1200

⊞ ⊞ ⊞ ⊞

☐

Debian Local Security Checks

0 of 18192

⊞ ⊞ ⊞ ⊞

☐

Default Accounts

0 of 315

⊞ ⊞ ⊞ ⊞

☒

Denial of Service

0 of 2404

⊞ ⊞ ⊞ ⊞

☐

FTP Local Security Checks

0 of 125

⊞ ⊞ ⊞ ⊞

☒

FTP

0 of 176

⊞ ⊞ ⊞ ⊞

☒

Fedora Local Security Checks

0 of 30209

⊞ ⊞ ⊞ ⊞

☐

FortiOS Local Security Checks

0 of 36

⊞ ⊞ ⊞ ⊞

☐

FreeBSD Local Security Checks

0 of 2009

⊞ ⊞ ⊞ ⊞

☐

Gain a shell remotely

0 of 109

⊞ ⊞ ⊞ ⊞

☐

General

0 of 8619

⊞ ⊞ ⊞ ⊞

☐

Gentoo Local Security Checks

0 of 2191

⊞ ⊞ ⊞ ⊞

☐

Cancel

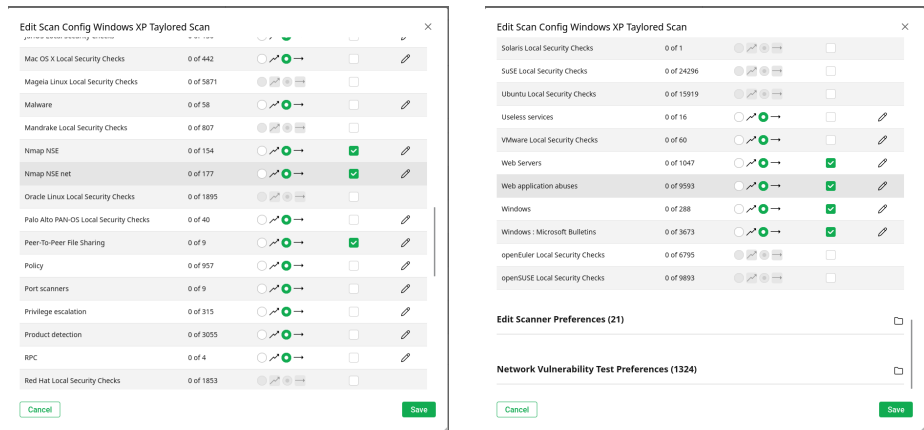
Save

(a) FTP & HTTP Taylored scan options

1

(b) FTP & HTTP Taylored scan options

2



(a) FTP & HTTP Taylored scan options
3

(b) FTP & HTTP Taylored scan options
4

Figure 1.13: Configuration for the FTP & HTTP taylored scan

This is far from being a completely optimized scan for our purpose, since we are able to better tailor each section seen here by pressing on the pencil mark, but that would become time consuming since you would select/deselect hundreds of different things it scans, so for simplicity it is ignored in this experiment.

1.1.3 Starting the scan

Now we are ready to begin the scan, to do this go back to the tasks windows, click on the task add button and add all the settings created so far like this:

New Task

Name: WIN XP FTP TASK

Comment:

Scan Targets: Windows XP FTP TARGET

Add results to Assets: ☒ Yes ☐ No

Apply Overrides: ☒ Yes ☐ No

Min QoD: 70

Alterable Task: ☒ Yes ☐ No

Auto Delete Reports: ☒ Do not automatically delete reports
☐ Automatically delete oldest reports but always keep newest (5 reports)

Scanner: OpenVAS Default

Scan Config: Full and fast

Cancel Save

(a) FTP task 1

New Task

Name:

Add results to Assets: ☒ Yes ☐ No

Apply Overrides: ☒ Yes ☐ No

Min QoD: 70

Alterable Task: ☒ Yes ☐ No

Auto Delete Reports: ☒ Do not automatically delete reports
☐ Automatically delete oldest reports but always keep newest (5 reports)

Scanner: OpenVAS Default

Scan Config: Full and fast

Order for target hosts: Sequential

Maximum concurrently executed NVTs per host: 4

Maximum concurrently scanned hosts: 20

Cancel Save

(b) FTP task 2

This task will scan Windows XP for the FTP service. Now you should have a new task created with the name you gave it, in this case **WIN XP FTP TASK** like so:



Now press the play button to queue it so that it will start, it will take approximately 10 minutes to complete.

New Task

Name: WIN XP HTTP&FTP TASK

Comment:

Scan Targets: WIN XP FTP&HTTP

Add results to Assets: ☒ Yes ☐ No

Apply Overrides: ☒ Yes ☐ No

Min QoD: 70

Alterable Task: ☒ Yes ☐ No

Auto Delete Reports: ☒ Do not automatically delete reports
☐ Automatically delete oldest reports but always keep newest (5 reports)

Scanner: OpenVAS Default

Scan Config: Windows XP Taylored Scan

Cancel Save

(a) FTP & HTTP Taylored task 1

New Task

Name:

Add results to Assets: ☒ Yes ☐ No

Apply Overrides: ☒ Yes ☐ No

Min QoD: 70

Alterable Task: ☒ Yes ☐ No

Auto Delete Reports: ☒ Do not automatically delete reports
☐ Automatically delete oldest reports but always keep newest (5 reports)

Scanner: OpenVAS Default

Scan Config: Windows XP Taylored Scan

Order for target hosts: Sequential

Maximum concurrently executed NVTs per host: 4

Maximum concurrently scanned hosts: 20

Cancel Save

(b) FTP & HTTP Taylored task 2

Chapter 2

AI Usage Declaration

This report was prepared with the assistance of AI tools for drafting and editing support. The structure and the core contents of the report were designed by the author. All final content was reviewed and approved by the author.